

OSPF DN-bit - пример на Cisco

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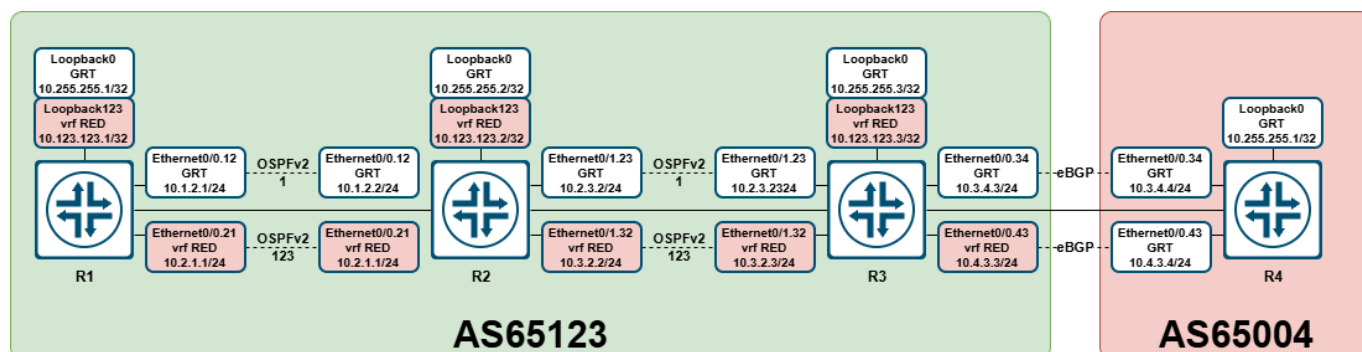
Краткое описание

В рамках данной заметки будет рассмотрено воспроизведение проблемы с DN-bit в OSPF.

DN-bit - это механизм в OSPF, позволяющий избежать петель маршрутизации при импортировании BGP-маршрутов в PE-CE взаимодействии в рамках MPLS L3VPN. Описан данный механизм в [RFC4576](#) и [RFC4577](#).

Если коротко и по простому: бывают такие ситуации, когда в процесс OSPF для какого-то VRF необходимо импортировать маршруты, полученные по BGP. И вот здесь может так случиться, что на соседнем роутере External LSA будет, а вот маршрута в RIB уже нет. Вот с этим и будем разбираться дальше 😊.

Топология



Конфигурация

R1:

```
vrf definition RED
 rd 10.255.255.1:123
 !
 address-family ipv4
 exit-address-family
 !
 interface Loopback0
 ip address 10.255.255.1 255.255.255.255
 ip ospf 1 area 0
 !
 interface Loopback123
```

```
vrf forwarding RED
ip address 10.123.123.1 255.255.255.255
ip ospf 123 area 0
!
interface Ethernet0/0
no ip address
!
interface Ethernet0/0.12
encapsulation dot1Q 12
ip address 10.1.2.1 255.255.255.0
ip ospf network point-to-point
ip ospf 1 area 0
!
interface Ethernet0/0.21
encapsulation dot1Q 21
vrf forwarding RED
ip address 10.2.1.1 255.255.255.0
ip ospf network point-to-point
ip ospf 123 area 0
!
router ospf 123 vrf RED
router-id 10.123.123.1
!
router ospf 1
router-id 10.255.255.1
!
```

R2:

```
vrf definition RED
rd 10.255.255.2:123
!
address-family ipv4
exit-address-family
!
interface Loopback0
ip address 10.255.255.2 255.255.255.255
ip ospf 1 area 0
!
interface Loopback123
vrf forwarding RED
ip address 10.123.123.2 255.255.255.255
ip ospf 123 area 0
!
interface Ethernet0/0
no ip address
!
interface Ethernet0/0.12
encapsulation dot1Q 12
ip address 10.1.2.2 255.255.255.0
ip ospf network point-to-point
ip ospf 1 area 0
```

```
!  
interface Ethernet0/0.21  
  encapsulation dot1Q 21  
  vrf forwarding RED  
  ip address 10.2.1.2 255.255.255.0  
  ip ospf network point-to-point  
  ip ospf 123 area 0  
!  
interface Ethernet0/1  
  no ip address  
!  
interface Ethernet0/1.23  
  encapsulation dot1Q 23  
  ip address 10.2.3.2 255.255.255.0  
  ip ospf network point-to-point  
  ip ospf 1 area 0  
!  
interface Ethernet0/1.32  
  encapsulation dot1Q 32  
  vrf forwarding RED  
  ip address 10.3.2.2 255.255.255.0  
  ip ospf network point-to-point  
  ip ospf 123 area 0  
!  
router ospf 123 vrf RED  
  router-id 10.123.123.2  
!  
router ospf 1  
  router-id 10.255.255.2  
!
```

R3:

```
vrf definition RED  
  rd 10.255.255.3:123  
  !  
  address-family ipv4  
  exit-address-family  
!  
interface Loopback0  
  ip address 10.255.255.3 255.255.255.255  
  ip ospf 1 area 0  
!  
interface Loopback123  
  vrf forwarding RED  
  ip address 10.123.123.3 255.255.255.255  
  ip ospf 123 area 0  
!  
interface Ethernet0/0  
  no ip address  
!  
interface Ethernet0/0.34
```

```
encapsulation dot1Q 34
ip address 10.3.4.3 255.255.255.0
!
interface Ethernet0/0.43
encapsulation dot1Q 43
vrf forwarding RED
ip address 10.4.3.3 255.255.255.0
!
interface Ethernet0/1
no ip address
!
interface Ethernet0/1.23
encapsulation dot1Q 23
ip address 10.2.3.3 255.255.255.0
ip ospf network point-to-point
ip ospf 1 area 0
!
interface Ethernet0/1.32
encapsulation dot1Q 32
vrf forwarding RED
ip address 10.3.2.3 255.255.255.0
ip ospf network point-to-point
ip ospf 123 area 0
!
router ospf 123 vrf RED
router-id 10.123.123.3
redistribute bgp 65123
!
router ospf 1
router-id 10.255.255.3
redistribute bgp 65123
!
router bgp 65123
bgp log-neighbor-changes
neighbor 10.3.4.4 remote-as 65004
!
address-family ipv4 vrf RED
neighbor 10.4.3.4 remote-as 65004
neighbor 10.4.3.4 activate
exit-address-family
!
```

R4:

```
interface Loopback0
ip address 10.255.255.4 255.255.255.255
!
interface Ethernet0/0
no ip address
!
interface Ethernet0/0.34
encapsulation dot1Q 34
```

```
ip address 10.3.4.4 255.255.255.0
!
interface Ethernet0/0.43
 encapsulation dot1Q 43
 ip address 10.4.3.4 255.255.255.0
!
router bgp 65004
 bgp log-neighbor-changes
 neighbor 10.3.4.3 remote-as 65123
 neighbor 10.4.3.3 remote-as 65123
!
 address-family ipv4
  network 10.255.255.4 mask 255.255.255.255
  neighbor 10.3.4.3 activate
  neighbor 10.4.3.3 activate
 exit-address-family
!
```

Как видно из конфигурации, между R3 и R4 подняты две eBGP-сессии. Со стороны R3 одна из них находится в vrf RED, вторая - в GRT. Также на R3 настроена редистрибуция из BGP в процессы OSPF.

Проверка

Проверяем таблицы маршрутизации на R1, R2, R3 как в GRT, так и в vrf RED.

Сначала в GRT:

R1:

```
R1#show ip route | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.1.2.0/24 is directly connected, Ethernet0/0.12
L       10.1.2.1/32 is directly connected, Ethernet0/0.12
O       10.2.3.0/24 [110/20] via 10.1.2.2, 00:28:24, Ethernet0/0.12
C       10.255.255.1/32 is directly connected, Loopback0
O       10.255.255.2/32 [110/11] via 10.1.2.2, 00:28:52, Ethernet0/0.12
O       10.255.255.3/32 [110/21] via 10.1.2.2, 00:26:35, Ethernet0/0.12
O E2    10.255.255.4/32 [110/1] via 10.1.2.2, 00:22:34, Ethernet0/0.12
R1#
```

R2:

```
R2#show ip route | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.1.2.0/24 is directly connected, Ethernet0/0.12
```

```
L      10.1.2.2/32 is directly connected, Ethernet0/0.12
C      10.2.3.0/24 is directly connected, Ethernet0/1.23
L      10.2.3.2/32 is directly connected, Ethernet0/1.23
O      10.255.255.1/32 [110/11] via 10.1.2.1, 00:28:59, Ethernet0/0.12
C      10.255.255.2/32 is directly connected, Loopback0
O      10.255.255.3/32 [110/11] via 10.2.3.3, 00:26:39, Ethernet0/1.23
O E2   10.255.255.4/32 [110/1] via 10.2.3.3, 00:22:33, Ethernet0/1.23
R2#
```

R3:

```
R3#show ip route | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O      10.1.2.0/24 [110/20] via 10.2.3.2, 00:26:39, Ethernet0/1.23
C      10.2.3.0/24 is directly connected, Ethernet0/1.23
L      10.2.3.3/32 is directly connected, Ethernet0/1.23
C      10.3.4.0/24 is directly connected, Ethernet0/0.34
L      10.3.4.3/32 is directly connected, Ethernet0/0.34
O      10.255.255.1/32 [110/21] via 10.2.3.2, 00:26:39, Ethernet0/1.23
O      10.255.255.2/32 [110/11] via 10.2.3.2, 00:26:39, Ethernet0/1.23
C      10.255.255.3/32 is directly connected, Loopback0
B      10.255.255.4/32 [20/0] via 10.3.4.4, 00:22:33
R3#
```

Как видно, на R1 и R2 попал маршрут до 10.255.255.4/32.

Теперь проверим в vrf RED:

R1:

```
R1#show ip route vrf RED | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C      10.2.1.0/24 is directly connected, Ethernet0/0.21
L      10.2.1.1/32 is directly connected, Ethernet0/0.21
O      10.3.2.0/24 [110/20] via 10.2.1.2, 00:03:47, Ethernet0/0.21
C      10.123.123.1/32 is directly connected, Loopback123
O      10.123.123.2/32 [110/11] via 10.2.1.2, 00:03:47, Ethernet0/0.21
O      10.123.123.3/32 [110/21] via 10.2.1.2, 00:03:47, Ethernet0/0.21
R1#
```

R2:

```
R2#show ip route vrf RED | b Gateway
Gateway of last resort is not set
```

```

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.2.1.0/24 is directly connected, Ethernet0/0.21
L       10.2.1.2/32 is directly connected, Ethernet0/0.21
C       10.3.2.0/24 is directly connected, Ethernet0/1.32
L       10.3.2.2/32 is directly connected, Ethernet0/1.32
O       10.123.123.1/32 [110/11] via 10.2.1.1, 00:03:47, Ethernet0/0.21
C       10.123.123.2/32 is directly connected, Loopback123
O       10.123.123.3/32 [110/11] via 10.3.2.3, 00:29:36, Ethernet0/1.32
R2#
```

R3:

```
R3#show ip route vrf RED | b Gateway
Gateway of last resort is not set
```

```

    10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.2.1.0/24 [110/20] via 10.3.2.2, 00:29:35, Ethernet0/1.32
C       10.3.2.0/24 is directly connected, Ethernet0/1.32
L       10.3.2.3/32 is directly connected, Ethernet0/1.32
C       10.4.3.0/24 is directly connected, Ethernet0/0.43
L       10.4.3.3/32 is directly connected, Ethernet0/0.43
O       10.123.123.1/32 [110/21] via 10.3.2.2, 00:03:47, Ethernet0/1.32
O       10.123.123.2/32 [110/11] via 10.3.2.2, 00:29:35, Ethernet0/1.32
C       10.123.123.3/32 is directly connected, Loopback123
B       10.255.255.4/32 [20/0] via 10.4.3.4, 00:25:53
R3#
```

В vrf RED ситуация отличается: на R1 и R2 нужный маршрут отсутствует. Проверим LSDB:

R1:

```
R1#show ip ospf 123 database
```

```
OSPF Router with ID (10.123.123.1) (Process ID 123)
```

```
Router Link States (Area 0)
```

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.123.123.1	10.123.123.1	328	0x80000007	0x00686C	3
10.123.123.2	10.123.123.2	329	0x80000009	0x004E2B	5
10.123.123.3	10.123.123.3	1756	0x80000004	0x000FBA	3

```
Type-5 AS External Link States
```

Link ID	ADV Router	Age	Seq#	Checksum	Tag
---------	------------	-----	------	----------	-----

```
10.255.255.4    10.123.123.3    1656    0x80000001 0x007677 3489726051
R1#
```

R2:

```
R2#show ip ospf 123 database
```

```
OSPF Router with ID (10.123.123.2) (Process ID 123)
```

```
Router Link States (Area 0)
```

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.123.123.1	10.123.123.1	329	0x80000007	0x00686C	3
10.123.123.2	10.123.123.2	328	0x80000009	0x004E2B	5
10.123.123.3	10.123.123.3	1755	0x80000004	0x000FBA	3

```
Type-5 AS External Link States
```

Link ID	ADV Router	Age	Seq#	Checksum	Tag
10.255.255.4	10.123.123.3	1656	0x80000001	0x007677	3489726051

```
R2#
```

R3:

```
R3#show ip ospf 123 database
```

```
OSPF Router with ID (10.123.123.3) (Process ID 123)
```

```
Router Link States (Area 0)
```

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.123.123.1	10.123.123.1	330	0x80000007	0x00686C	3
10.123.123.2	10.123.123.2	329	0x80000009	0x004E2B	5
10.123.123.3	10.123.123.3	1754	0x80000004	0x000FBA	3

```
Type-5 AS External Link States
```

Link ID	ADV Router	Age	Seq#	Checksum	Tag
10.255.255.4	10.123.123.3	1655	0x80000001	0x007677	3489726051

```
R3#
```

Как видно, LSA Type 5 для префикса 10.255.255.4/32 внутри процесса для vrf была выпущена. Посмотрим на нее внимательней:

R2:


```
R2#show ip ospf 123 database external 10.255.255.4

      OSPF Router with ID (10.123.123.2) (Process ID 123)

      Type-5 AS External Link States

LS age: 1724
Options: (No TOS-capability, DC, Downward)
LS Type: AS External Link
Link State ID: 10.255.255.4 (External Network Number )
Advertising Router: 10.123.123.3
LS Seq Number: 80000001
Checksum: 0x7677
Length: 36
Network Mask: /32
    Metric Type: 2 (Larger than any link state path)
    MTID: 0
    Metric: 1
    Forward Address: 0.0.0.0
    External Route Tag: 3489726051
```

R2#

Сравним с аналогичной LSA, но в GRT:

R2:

```
R2#show ip ospf 1 database external 10.255.255.4

      OSPF Router with ID (10.255.255.2) (Process ID 1)

      Type-5 AS External Link States

LS age: 1807
Options: (No TOS-capability, DC, Upward)
LS Type: AS External Link
Link State ID: 10.255.255.4 (External Network Number )
Advertising Router: 10.255.255.3
LS Seq Number: 80000001
Checksum: 0x9815
Length: 36
Network Mask: /32
    Metric Type: 2 (Larger than any link state path)
    MTID: 0
    Metric: 1
    Forward Address: 0.0.0.0
    External Route Tag: 65004
```

R2#

Также можно посмотреть на эти LSA из дампа:

GRT LSA 5

```
> Frame 73: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: aa:bb:cc:00:30:10, Dst: 01:00:5e:00:00:05
> 802.1Q Virtual LAN, PVID: 0, DEI: 0, ID: 23
> Internet Protocol Version 4, Src: 10.2.3.3, Dst: 224.0.0.5
> Open Shortest Path First
  > OSPF Header
    > LS Update Packet
      Number of LSAs: 1
      > LSA-type 5 (AS-External-LSA (ASBR)), len 36
        .000 0000 0000 0001 = LS Age (seconds): 1
        0... .. = Do Not Age Flag: 0
      > Options: 0x20, (DC) Demand Circuits
      LS Type: AS-External-LSA (ASBR) (5)
      Link State ID: 10.255.255.4
      Advertising Router: 10.255.255.3
      Sequence Number: 0x00000002
      Checksum: 0x9016
      Length: 36
      Netmask: 255.255.255.255
      1... .. = External Type: Type 2 (metric is larger than any other link state path)
        .000 0000 = TOS: 0
      Metric: 1
      Forwarding Address: 0.0.0.0
      External Route Tag: 65064
```

VRF LSA 5

```
> Frame 74: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: aa:bb:cc:00:30:10, Dst: 01:00:5e:00:00:05
> 802.1Q Virtual LAN, PVID: 0, DEI: 0, ID: 33
> Internet Protocol Version 4, Src: 10.3.2.3, Dst: 224.0.0.5
> Open Shortest Path First
  > OSPF Header
    > LS Update Packet
      Number of LSAs: 1
      > LSA-type 5 (AS-External-LSA (ASBR)), len 36
        .000 0000 0000 0001 = LS Age (seconds): 1
        0... .. = Do Not Age Flag: 0
      > Options: 0x00, DN, (DC) Demand Circuits
      LS Type: AS-External-LSA (ASBR) (5)
      Link State ID: 10.255.255.4
      Advertising Router: 10.123.123.3
      Sequence Number: 0x00000002
      Checksum: 0x7478
      Length: 36
      Netmask: 255.255.255.255
      1... .. = External Type: Type 2 (metric is larger than any other link state path)
        .000 0000 = TOS: 0
      Metric: 1
      Forwarding Address: 0.0.0.0
      External Route Tag: 3489726951
```

Проблема в DN-бите, который проставляет R3 при редистрибуции из BGP в OSPF в vrf RED. Этот механизм используется для того, чтобы избежать появления петель (см. [RFC4576](#) и [RFC4577](#)). Если коротко, то маршрутизатор обязан игнорировать LSA с проставленным DN-битом и не использовать информацию из таких LSA во время расчета маршрутов в OSPF.

Для проверки настроим на R1 процесс BGP:

R1:

```
!
router bgp 65123
  bgp log-neighbor-changes
!
  address-family ipv4
    redistribute ospf 1 match internal external 1 external 2
  exit-address-family
!
  address-family ipv4 vrf RED
    redistribute ospf 123 match internal external 1 external 2
  exit-address-family
!
```

Ремарка: указание в явном виде, что импортируются как внутренние, так и внешние маршруты, т.к. Cisco по умолчанию не импортирует external-маршруты.

Теперь проверим:

R1:

```
R1#show bgp ipv4 unicast | b Network
```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	10.1.2.0/24	0.0.0.0	0		32768	?
*>	10.2.3.0/24	10.1.2.2	20		32768	?
*>	10.255.255.1/32	0.0.0.0	0		32768	?
*>	10.255.255.2/32	10.1.2.2	11		32768	?
*>	10.255.255.3/32	10.1.2.2	21		32768	?
*>	10.255.255.4/32	10.1.2.2	1		32768	?

```
R1#show bgp vpnv4 unicast vrf RED | b Network
      Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 10.255.255.1:123 (default for vrf RED)
*>  10.2.1.0/24        0.0.0.0              0          32768 ?
*>  10.3.2.0/24        10.2.1.2             20         32768 ?
*>  10.123.123.1/32    0.0.0.0              0          32768 ?
*>  10.123.123.2/32    10.2.1.2             11         32768 ?
*>  10.123.123.3/32    10.2.1.2             21         32768 ?
R1#
```

Как решить эту проблему? В случае Cisco эта проблема решается добавлением в конфигурацию инструкции **capability vrf-lite** в контексте процесса OSPF для vrf (у других вендоров тоже есть аналогичные решения, у Huawei, например, это **vpn-instance-capability simple**).

Проверим на R1:

R1:

```
R1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 123 vrf RED
R1(config-router)#capability vrf-lite
R1(config-router)#
*Mar 26 19:19:23.243: %OSPF-5-ADJCHG: Process 123, Nbr 10.123.123.2 on
Ethernet0/0.21 from FULL to DOWN, Neighbor Down: Interface down or detached
*Mar 26 19:19:23.245: %OSPF-5-ADJCHG: Process 123, Nbr 10.123.123.2 on
Ethernet0/0.21 from LOADING to FULL, Loading Done
R1(config-router)#end
R1#
*Mar 26 19:19:36.055: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.1.2.0/24 is directly connected, Ethernet0/0.12
L       10.1.2.1/32 is directly connected, Ethernet0/0.12
O       10.2.3.0/24 [110/20] via 10.1.2.2, 00:07:51, Ethernet0/0.12
C       10.255.255.1/32 is directly connected, Loopback0
O       10.255.255.2/32 [110/11] via 10.1.2.2, 00:07:51, Ethernet0/0.12
O       10.255.255.3/32 [110/21] via 10.1.2.2, 00:07:51, Ethernet0/0.12
O E2    10.255.255.4/32 [110/1] via 10.1.2.2, 00:07:51, Ethernet0/0.12
R1#show ip route vrf RED | b Gateway
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.2.1.0/24 is directly connected, Ethernet0/0.21
L       10.2.1.1/32 is directly connected, Ethernet0/0.21
O       10.3.2.0/24 [110/20] via 10.2.1.2, 00:00:25, Ethernet0/0.21
C       10.123.123.1/32 is directly connected, Loopback123
O       10.123.123.2/32 [110/11] via 10.2.1.2, 00:00:25, Ethernet0/0.21
```

```

O      10.123.123.3/32 [110/21] via 10.2.1.2, 00:00:25, Ethernet0/0.21
O E2    10.255.255.4/32 [110/1] via 10.2.1.2, 00:00:25, Ethernet0/0.21
R1#show bgp ipv4 unicast | b Network
      Network          Next Hop          Metric LocPrf Weight Path
*>  10.1.2.0/24        0.0.0.0              0         32768 ?
*>  10.2.3.0/24        10.1.2.2             20        32768 ?
*>  10.255.255.1/32    0.0.0.0              0         32768 ?
*>  10.255.255.2/32    10.1.2.2             11        32768 ?
*>  10.255.255.3/32    10.1.2.2             21        32768 ?
*>  10.255.255.4/32    10.1.2.2             1         32768 ?
R1#show bgp vpnv4 unicast vrf RED | b Network
      Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 10.255.255.1:123 (default for vrf RED)
*>  10.2.1.0/24        0.0.0.0              0         32768 ?
*>  10.3.2.0/24        10.2.1.2             20        32768 ?
*>  10.123.123.1/32    0.0.0.0              0         32768 ?
*>  10.123.123.2/32    10.2.1.2             11        32768 ?
*>  10.123.123.3/32    10.2.1.2             21        32768 ?
*>  10.255.255.4/32    10.2.1.2             1         32768 i
R1#

```

На всякий случай уточню, что данная инструкция является локальной, и на R2 ничего не изменилось:

R2:

```

R2#
*Mar 26 19:19:27.888: %OSPF-5-ADJCHG: Process 123, Nbr 10.123.123.1 on
Ethernet0/0.21 from LOADING to FULL, Loading Done
R2#show ip route vrf RED | b Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C      10.2.1.0/24 is directly connected, Ethernet0/0.21
L      10.2.1.2/32 is directly connected, Ethernet0/0.21
C      10.3.2.0/24 is directly connected, Ethernet0/1.32
L      10.3.2.2/32 is directly connected, Ethernet0/1.32
O      10.123.123.1/32 [110/11] via 10.2.1.1, 00:02:01, Ethernet0/0.21
C      10.123.123.2/32 is directly connected, Loopback123
O      10.123.123.3/32 [110/11] via 10.3.2.3, 00:47:32, Ethernet0/1.32
R2#

```